Cradero

Complete Maintenance Management

13	3	100+	4
MODULES	TIERS	FEATURES	USER ROLES

A modern, cloud-based CMMS built for manufacturing, food production, and facilities teams. Manage work orders, equipment, inventory, compliance, and analytics from a single platform.

Basic Tier — Core Modules

Included in all tiers — the foundation of your maintenance operations

Work Order Management

Create, assign, track, and complete work orders with priority levels, photo attachments, time tracking, comments, and parts usage logging. Full lifecycle management from open to completed.

Equipment & Asset Register

Comprehensive equipment database with manufacturer details, location, install dates, criticality ratings (A/B/C), warranty tracking, maintenance history, and health scoring.

Preventive Maintenance

Schedule recurring PM tasks with customisable checklists, frequency-based auto-scheduling, calendar sync, completion workflows, and automatic next-due-date calculation.

Reactive Maintenance

Log breakdowns with fault cause classification, root cause analysis, downtime tracking, engineer time recording, and optional auto-generation of linked work orders.

Work Requests

Production staff can submit maintenance requests with urgency levels. Managers review and approve requests, automatically converting them into assigned work orders.

Three-Tier Access

Basic — 5 core modules for essential maintenance operations. **Essential** — adds Notifications, Daily Log, My Tasks, Audit Trail, Permits, and CSV Import. **Pro** — unlocks everything including Parts Inventory, Purchase Orders, Analytics, Calendar, QR codes, dark mode, and offline access.

Essential & Pro Modules

Additional capabilities beyond Basic tier

Essential Tier Modules

Notifications & Alerts ESSENTIAL

Dynamic notification bell showing overdue PMs, unassigned work orders, high-priority tasks, low-stock parts, and equipment-down alerts.

Daily Activity Log ESSENTIAL

Timeline view of all maintenance activities for any selected date. Visual activity feed with completion tracking and daily summary statistics.

Engineer Dashboard (My Tasks) ESSENTIAL

Personalised task view showing assigned work orders and PM tasks sorted by priority. Quick-start functionality, today's summary stats, and overdue task highlighting.

Audit Trail ESSENTIAL

Every create, update, delete, and complete action is logged with user identity, timestamp, entity type, and change details. Filterable by action, entity, and date.

Permits Management ESSENTIAL

Issue and manage hot work permits and contractor work permits with safety checklists, compliance verification, expiry tracking, and a closed permits archive for auditing.

CSV Import Wizard ESSENTIAL

4-step import wizard: upload file, map columns with auto-detection, preview with validation and duplicate detection, then batch import. Supports equipment, parts, and PM schedules.

Pro Tier Modules

Parts Inventory PRO

Track spare parts with stock levels, minimum thresholds, shelf/bin locations, supplier information, compatible equipment linking, and automated low-stock alerts.

Purchase Orders PRO

Create, approve, and track purchase orders with vendor management, line items, budget codes, priority shipping, and PDF export. Full procurement workflow from request to receipt.

Analytics & Reports PRO

Interactive charts and KPIs including equipment downtime, WO status distribution, PM compliance trends, MTTR/MTBF, fault cause analysis, technician workload, and cost breakdowns.

Maintenance Calendar PRO

Visual month and week views with colour-coded event types. Plan weekly PMs, monthly services, major shutdowns, and inspections. Auto-syncs with PM schedule.

Analytics Detail

- Equipment downtime by line (bar chart)
- Work order status distribution (doughnut chart)
- PM compliance trend (6-month line chart)
- Top 5 problem equipment ranking
- MTTR & MTBF per equipment
- Fault cause frequency analysis
- Recurring fault pattern detection
- Technician workload overview
- Equipment health scores (0-100)
- Labor cost calculation by engineer
- Downtime hours by equipment
- Engineer hours tracking
- Downtime & engineer hours trends (6-month)
- Parts stock depletion forecast
- Maintenance cost breakdown
- PM completion rate statistics

PDF Report Export

Export individual data sets (work orders, equipment, parts, breakdowns, PM schedule, purchase orders) as formatted PDFs, or generate a comprehensive full maintenance report covering all KPIs, equipment status, active work orders, PM schedule, and parts requiring attention.

Feature Detail

Deep dive into key workflows

Work Order Lifecycle

- Create with equipment, title, description, and priority
- Assign to engineers from team profiles
- Attach photos during creation or later
- Track start time, finish time, and total job duration
- Record machine downtime hours
- Log parts used with automatic stock deduction
- Return parts to stores if removed in error
- Add timestamped comments for communication
- View linked equipment details and history
- Filter by status, priority, or sort order
- Full-text search across all work order fields
- Status flow: Open → In Progress → Completed / On Hold

Preventive Maintenance Workflow

- Define PM tasks with custom checklists
- Set frequency: daily, weekly, bi-weekly, monthly, quarterly, semi-annual, annual
- Assign to engineers with reassignment capability
- Automatic due date calculation and status tracking
- Checklist-driven completion — all items must be ticked
- Record start/finish times during completion
- Log parts used during PM with stock deduction
- Attach photos during completion (managers/admins)
- Raise follow-up work orders for issues found
- Auto-calculate next due date after completion
- Auto-sync calendar events with PM schedule
- Edit PM tasks (managers/admins) with full checklist management
- View completion history with checklist results and notes
- Filter by status, equipment, and date range

Reactive Maintenance

- 10 fault cause categories: electrical, mechanical, hydraulic, pneumatic, software/PLC, operator error, wear & tear, contamination, calibration, other
- Downtime start/end tracking with auto-calculation
- Engineer job start/finish tracking
- Root cause detail field for deeper analysis
- Resolution/fix description
- Auto-raise linked work order option
- Filter by machine, status, fault cause, and date range
- Edit all fields after initial logging
- Auto-close linked WO when breakdown is resolved

Equipment, Inventory & Security

Asset management, stock control, and access governance

Equipment Management

- Full asset register with name, ID, manufacturer, model
- Location tracking per equipment
- Criticality ratings: A (Critical), B (Important), C (Low)
- Warranty tracking with expiry alerts
- Equipment health scoring (0-100) based on age, breakdowns, status, and WO load
- Downtime analysis per equipment with MTTR, MTBF, and availability %
- Compatible parts linking
- SOP document attachments (PDF, Word, images)
- Manufacturer manual attachments
- QR code generation and scanning for quick lookup
- Work order history (30-day count)
- CSV bulk import with column mapping wizard

Parts & Inventory

- Part name, number, quantity, min stock, shelf/bin location
- Multi-supplier tracking with pricing and lead times
- Compatible equipment linking via checkboxes
- Automatic stock deduction on WO/PM part usage
- Return-to-stores functionality
- Low stock and out-of-stock alerts in notifications
- Stock forecast: months until depletion
- Searchable parts dropdown with live stock display
- Stock warnings when selecting parts for usage
- CSV bulk import for inventory migration

Security & Access Control

Authentication

Supabase-powered authentication with email/password login, account registration with

Session Security

30-minute inactivity timeout with 5-minute warning. Client-side rate limiting (5 attempts,

admin approval workflow, and secure session management.

30s lockout). Automatic session expiry on tab visibility change.

Content Security Policy

Strict CSP headers limiting script sources, style sources, image sources, and connection endpoints. Frame ancestors blocked to prevent clickjacking.

Input Validation

Comprehensive input sanitisation with length limits, HTML escaping, file type validation, size limits (10MB), and suspicious double-extension detection.

Roles, Tiers & Platform

Access control, pricing tiers, and technical capabilities

User Roles

Admin

Full access to all features. Manage users, approve accounts, delete records, edit equipment, and configure system settings.

Manager

Full operational access. Create/edit/delete work orders, equipment, and PM tasks. Approve work requests. Edit PM checklists. Upload SOPs.

Engineer

Work on assigned tasks. Complete work orders and PM checklists. Log breakdowns. Record time and parts usage. Submit comments.

Production

Submit maintenance requests via the Work Requests tab. View request status. Limited to request submission — no access to operational modules.

Tier Comparison

Feature	Basic	Essential	Pro
Work Order Management	✓	✓	✓
Equipment Register	✓	✓	✓
Preventive Maintenance	✓	✓	✓
Reactive Maintenance	✓	✓	✓
Work Requests	✓	✓	✓
Notifications & Alerts	—	✓	✓
CSV Import	—	✓	✓

Daily Activity Log	—	✓	✓
Engineer Dashboard (My Tasks)	—	✓	✓
Audit Trail	—	✓	✓
Permits (Hot Work & Contractor)	—	✓	✓
Parts Inventory	—	—	✓
Purchase Orders	—	—	✓
Analytics & Reports	—	—	✓
Maintenance Calendar	—	—	✓
QR Code Generation & Scanning	—	—	✓
Dark Mode	—	—	✓
SOP & Manual Attachments	—	—	✓
Stock Forecast & Supplier Management	—	—	✓
Equipment Health Scores	—	—	✓
Offline Mode (Service Worker)	—	—	✓

Platform & Technical

Architecture, integrations, and deployment

Platform Features

Cloud-Based

Hosted on Supabase with PostgreSQL database, real-time subscriptions, row-level security, and managed authentication. No servers to maintain.

Mobile Responsive

Fully responsive design with slide-out navigation drawer, touch-friendly 44px tap targets, full-screen modals on mobile, and horizontal scroll for data tables.

Real-Time Notifications

Dynamic notification bell showing overdue PMs, unassigned work orders, high-priority tasks, low-stock parts, and equipment-down alerts.

Full Audit Trail

Every create, update, delete, and complete action is logged with user identity, timestamp, entity type, and change details. Filterable by action, entity, and date.

Photo Management

Upload photos to work orders during creation or on the detail view. Stored securely in Supabase Storage. Drag-and-drop support with file type and size validation.

CSV Import Wizard

4-step import wizard: upload file, map columns with auto-detection, preview with validation and duplicate detection, then batch import. Supports equipment, parts, and PM schedules.

PDF Generation

Generate professional PDF reports for purchase orders and full maintenance reports using jsPDF. Includes headers, tables, KPIs, and page numbering.

QR Codes PRO

Generate scannable QR codes for any equipment using qrcode-generator. Scan via device camera using jsQR for instant equipment lookup. Manual ID entry fallback.

Technical Stack

Frontend

Single-page application built with vanilla JavaScript. No framework dependencies. Work

Sans + IBM Plex Mono typography. CSS custom properties for theming.

Backend

Supabase (PostgreSQL + Auth + Storage + Row Level Security). RESTful data access layer with

typed API wrapper. Real-time capabilities via Supabase subscriptions.

Charts & Visualisation

Chart.js 4.4 for interactive bar, doughnut, line, and trend charts. Responsive with maintained aspect ratios. Dynamic data-driven rendering.

Libraries

Supabase JS SDK v2.97, Chart.js v4.4, jsPDF v2.5 (PDF generation), jsQR v1.4 (QR scanning), qrcode-generator v1.4. All loaded via CDN with SRI hashes.

Single-File Architecture

Cradero is delivered as a single HTML file (plus a config.js for credentials), making deployment as simple as hosting a static file. No build step, no bundler, no npm install. The entire application — HTML structure, CSS styles, and JavaScript logic — lives in one file for maximum portability and zero-config deployment.